

Day 3: Tasks

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.



Parameter	Value
IPv4 Address	: 192.168.56.1
Subnet Mask	: 255.255.255.0
Default Gateway	: 192.168.31.1

3. **Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.**
Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.



Current Network Information

Setting	Value
Router IP (Default Gateway)	192.168.31.1
Subnet Mask	255.255.255.0
Static IP Assigned	192.168.31.50

Procedure

1. Opened Network and Sharing Center and selected the active Wi-Fi connection.
2. Opened Internet Protocol Version 4 (TCP/IPv4) Properties.
3. Changed the setting from Obtain an IP address automatically (DHCP) to Use the following IP address.
4. Entered the following values:
 - o IP Address: 192.168.31.50
 - o Subnet Mask: 255.255.255.0
 - o Default Gateway: 192.168.31.1
 - o Preferred DNS Server: 8.8.8.8
5. Saved the settings and tested the internet connection by opening websites in a browser.

Observation

The internet connection remained active after assigning the static IP address, confirming that the configuration was correct.

Reverting to DHCP

After testing, the settings were changed back to:

- Obtain an IP address automatically
- Obtain DNS server address automatically

4. **In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.**



DHCP Range Information

In the router's admin panel, the DHCP settings were located and the current DHCP range was found as:

- Start IP Address: 192.168.31.100
 - End IP Address: 192.168.31.200
- (The actual range may vary depending on the router configuration.)

What Happens If Two Devices Have the Same IP Address?

If two devices are assigned the same IP address, an IP address conflict occurs. As a result, one or both devices may lose network connectivity, experience unstable internet access, or be unable to communicate properly on the network. This can cause interruptions in browsing, file sharing, and other network services.